



Available Reliable Accountable

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PowerCombo Datasheet

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PowerCombo®

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PowerCombo®

PowerCombo, a high-performance, all-in-one, containerized battery energy storage system developed by UNICOM BSS, provides large-scale applications with intelligent and reliable BESS for auxiliary services etc. As a leading BESS product, PowerCombo is certificated by UL9540, UL1973, UL9540A, IEC62619, IEC 62933, IEC 62477, CE MARK, UN3536, NFPA855, bankability report, provides secure, reliable and safe power supply.

- Auxiliary Service
- Behind-The-Meter
- Front-The-Meter
- Maximum Demand Management
- Peak Shaving
- EV Charging Power Supplyment

CE Mark by TUV

Bankability Report by DNV

UL9540

UL9540A

UN38.3

UN3536

NFPA855

IEC62933

IEEE1547

IEC62619

UL1973

IEC62477

UL1741

EN50549

G99



Smart and User-friendly

Frequency / Voltage Stability;
Grid Following / Grid Forming;
Cloud dispatching and operation data analysis;



Economical and High Availability

Supports peak shaving, ADR, FCR/aFRR applications and more;
Supports remote software update and components upgrade,
lower operation and maintenance costs;



Safe and Reliable

Fault processing mechanism, responds to
multiple fault scenarios;
Customized BMS provides comprehensive
measurement and protection functions;



Flexible and Convenient

Modular design, single unit management;
Plug-and-play, supporting zero-test on site.

➤ PowerCombo-20C2H1600K



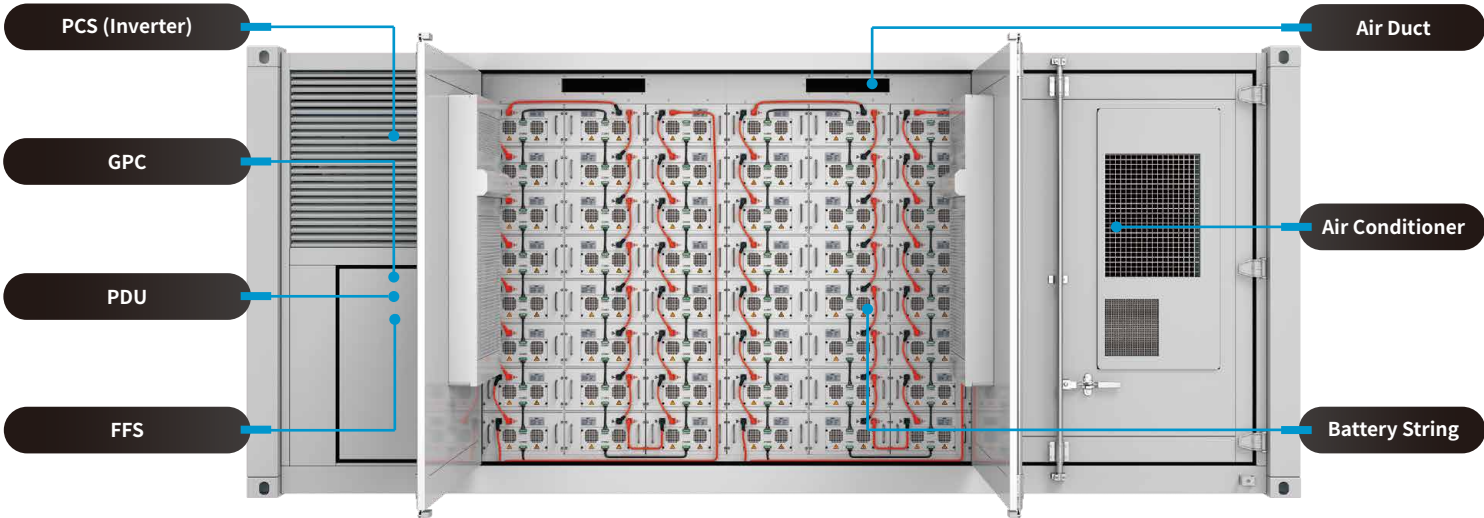
20C2H1600K-8LS385-8LP48



System Technical Specifications

Item	20C2H1600K	20C2H1000KLS
DC Data		
Battery chemistry	Lithium Iron Phosphate (LFP)	
Cell life cycle	8,000 cycles with 70% retention @0.5C 25°C	
Cell spec	3.2V/314Ah	
String configuration	1P384S	1P288S
Number of strings	8	8
DC rated energy capacity	3,086kWh	2,315kWh
Rated voltage	1,228.8V	921.6V
Voltage range	1,075.2V~1,363.2V	806.4V~1,022.4V
BMS communication interface	RS485, Ethernet	
BMS communication protocol	Modbus RTU, Modbus TCP	
AC Data	Standard Output	
Rated AC Power	1,600kW	1,000kW
Maximum AC power	1,760kW	1,100kW
Rated voltage	690V	400V
Grid voltage range	586.5~759V(Optional)	340~440V(Optional)
AC rate of current	1,338.8A	1,443.4A
Output THDi	<3%	
AC PF	-1~+1	
AC output	3W+PE	
General Data		
Dimension w/o clearances (L*W*H)	6,058x2,438x2,896mm	
Weight of the whole system	≤31t	≤26.1t
Degree of protection	IP54	
Operating temperature range	-30~50°C	
Relative humidity	0~95% (non-condensing)	
Max working altitude	2,000m/6,562feet(non-derating)	
Cooling concept of DC hatch	Liquid Cooling and Dehumidification system	
Fire fighting system	FK-5-1-12/Water fire fighting	
Communication interfaces	RS485, Ethernet	
Certificates	UL9540, IEC62933, UN3536, CE MARK by TÜV Rheinland	

➤ PowerCombo-20C2H1200K



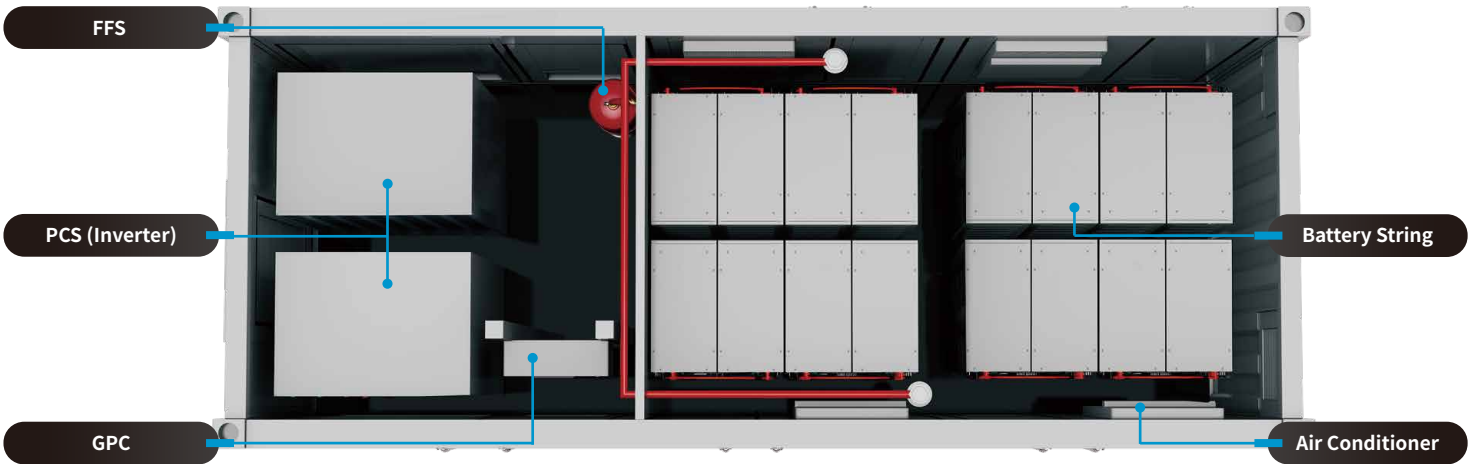
20C2H1200K-6S418-24P16



System Technical Specifications

Item	20C2H1200K
DC Data	
Battery chemistry	Lithium Iron Phosphate (LFP)
Cell life cycle	8,000 cycles with 70% retention @ 0.5C 25°C
Cell spec	3.2V/314Ah
String configuration	1P384S
Number of strings	6
DC rated energy capacity	2,315kWh
Rated voltage	1,228.8V
Voltage range	1,075.2V~1,363.2V
BMS communication interface	RS485, Ethernet
BMS communication protocol	Modbus RTU, Modbus TCP
AC Data	
Rated AC Power	1,200kW
Maximum AC power	1,320kW
Rated voltage	690V
Grid voltage range	586.5~759V(configurable)
AC rate of current	1,004.1A
Output THDi	<3%
AC PF	-1~+1
AC output	3W+PE
General Data	
Dimension w/o clearances (W*D*H)	6,058x2,438x2,591mm
Weight of the whole system	26.5t
Degree of protection	IP54
Operating temperature range	-35~40°C
Relative humidity	0~95% (non-condensing)
Max working altitude	2,000m/6,562feet (non-derating)
Cooling concept of DC hatch	Air cooling
Fire fighting system	FK-5-1-12/Water fire fighting
Communication interfaces	RS485, Ethernet
Certificates	UL9540, IEC 62933, UN3536, CE Mark by TÜV

➤ PowerCombo-20C1H1000K



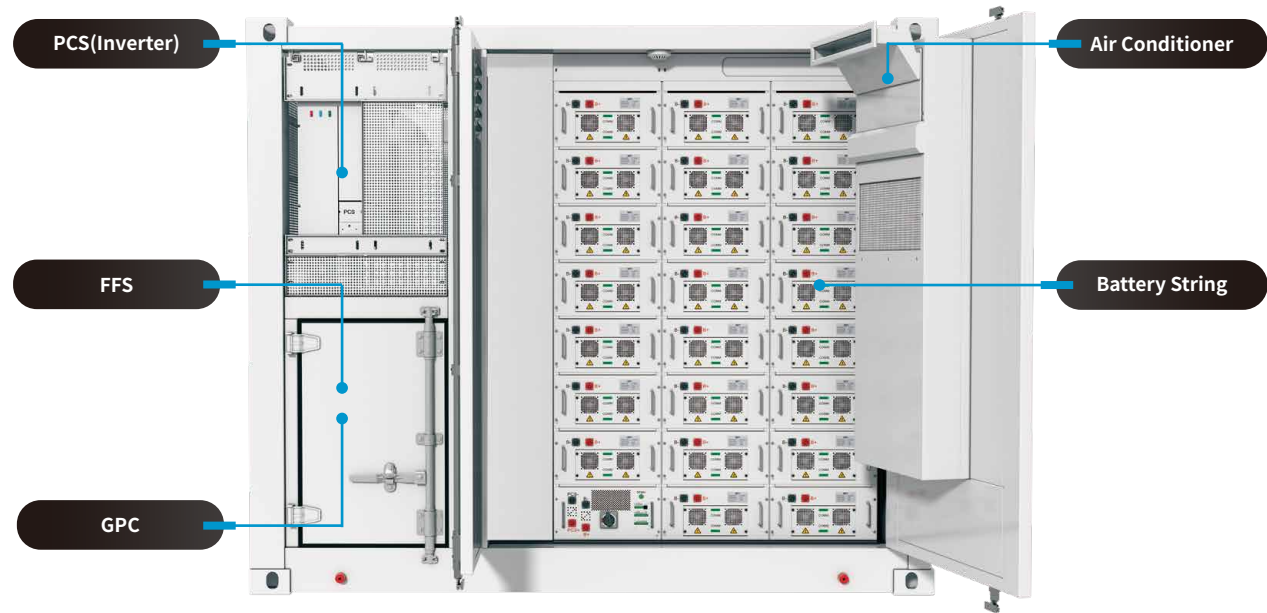
20C1H1000K-8S138-15P9



System Technical Specifications

Item	20C1H1000K
DC Data	
Battery chemistry	Lithium Iron Phosphate (LFP)
Cell life cycle	6,000 cycles with 80% Retention @1C 25°C
Cell spec	3.2V/90Ah
String configuration	2P240S
Number of strings	8
DC rated energy capacity	1,105kWh
Rated voltage	768V
Voltage range	672V~852V
BMS communication interface	RS485, Ethernet
BMS communication protocol	Modbus RTU, Modbus TCP
AC Data	
Rated AC power	1000kW
Maximum AC power	1100kW
Rated voltage	400V
Grid voltage range	360~440V (configurable)
AC rate of current	1443.4A
Output THDi	<3%
AC PF	-1~+1
AC output	3W+PE
General Data	
Dimension w/o clearances (W*D*H)	6,058x2,438x2,591mm
Weight of the whole system	19.5t
Degree of protection	IP54
Operating temperature range	-30~55°C
Relative humidity	0~95% (non-condensing)
Max working altitude	2,000m/6,562feet (non-derating)
Cooling concept of DC hatch	Air cooling
Fire fighting system	FK-5-1-12/Water fire fighting
Communication interfaces	RS485, Ethernet
Certificates	UL9540, UN3536

➤ PowerCombo-10C2H360K



10C2H360K-2S418-23P16

Container Size Minimum Discharge Hour AC Power Quantity of Battery String Code of Battery String Quantity of Pack per String 16kWh/Pack

System Technical Specifications

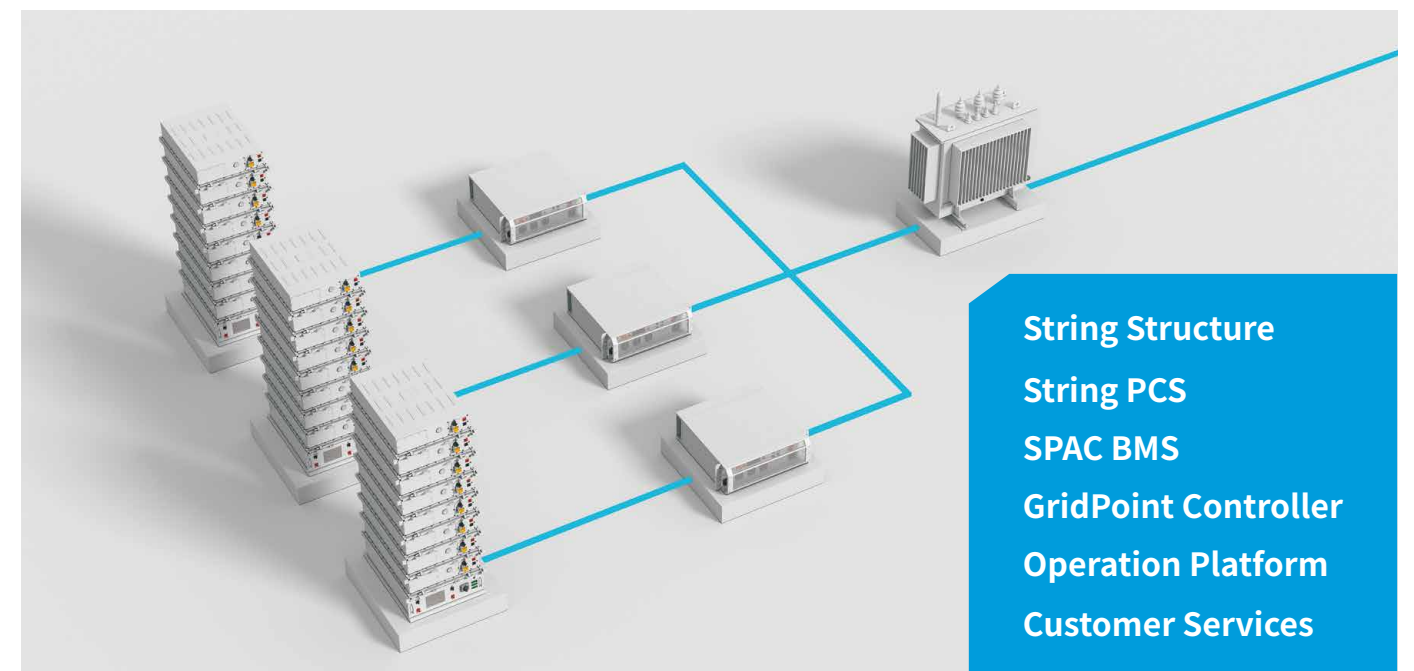
Item	10C2H250K	10C2H360K
DC Data		
Battery chemistry	Lithium Iron Phosphate (LFP)	
Cell life cycle	8,000 cycles with 70% retention @0.5C 25°C	
Cell spec	3.2V/314Ah	
String configuration	1P320S	1P368S
Number of strings	2	
Rated energy capacity	643kWh	740kWh
Rated voltage	1,024V	1,177.6V
Voltage range	896V~1,136V	1,030.4V~1,306.4V
BMS communication interface	RS485, Ethernet	
BMS communication protocol	Modbus RTU, Modbus TCP	
AC Data		
Rated AC power	250kW	360kW
Maximum AC power	275kW	396kW
Rated voltage	400V	690V
Grid voltage range	-15%~+10% (configurable)	
AC rate of current	360.9A	301.2A
Output THDi	<3%	
AC PF	-1~+1	
AC output	3W+PE	
General Data		
Dimension(L*W*H)	2,991x2,438x2,591mm	
Weight of the whole system	9.2t	9.9t
Degree of protection	IP54	
Operating temperature range	-30~55°C	
Relative humidity	0~95% (non-condensing)	
Max working altitude	2,000m/6,562feet (non-derating)	
Cooling concept of DC hatch	Air cooling	
Fire fighting system	FK-5-1-12/Water fire fighting	
Communication interfaces	RS485, Ethernet	
Certificates	UL9540, UN3536	



Advanced Technologies

UNICOM not only leads in system integration but also has extraordinary capacities in innovation and developing leading BESS technologies.

By virtue of self-developed String Structure BESS, SPAC BMS and O&M System, UNICOM provides products with high energy density, intelligence, availability and low fault rate. Maximizing the project revenue and guaranteeing the return on investment under the premise of safe and stable operation of BESS projects, acquiring more value and revenue from the BESS deployment.



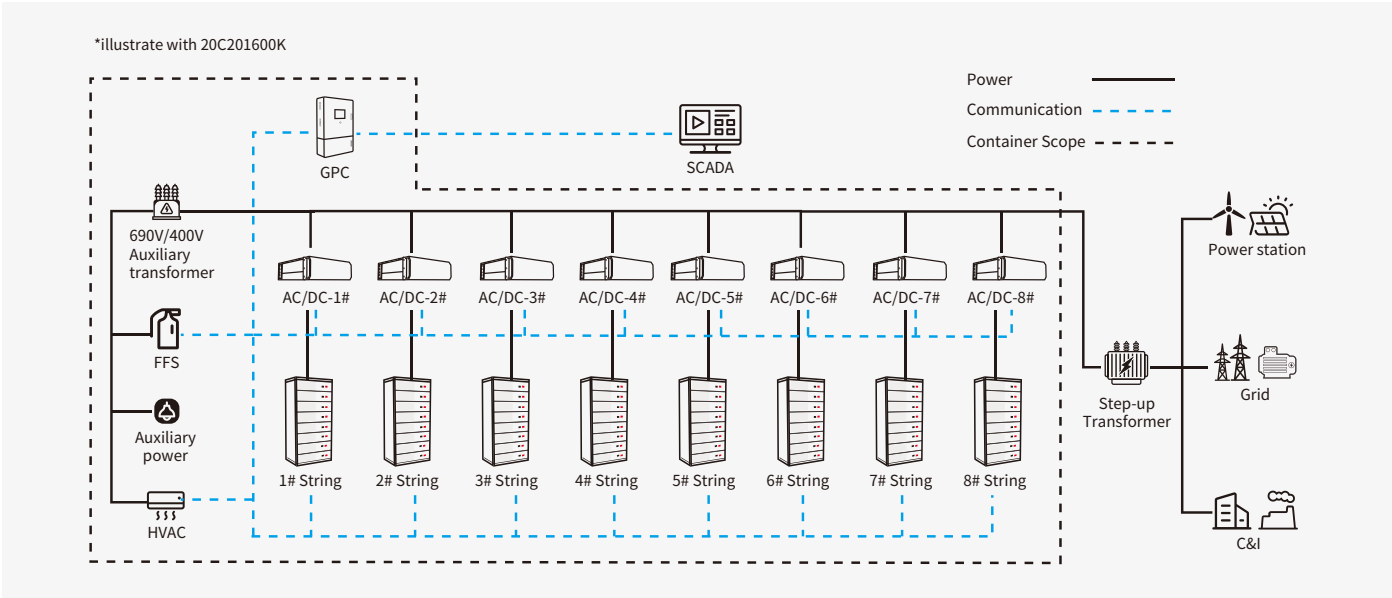
String Structure

String Structure BESS is a system design that leads each battery string to connect to its own modular PCS individually. Under this topology, each string is an independent unit-structure in DC circuitry. Charging and discharging of each battery string are independent, leads to smaller short-circuit current, and branch faults do not affect the operation of the entire BESS.

Features

- Guaranteeing High Availability
- Long System Lifespan
- Simple O&M

Topology



Advantages

One-on-one Design

Extreme Safety

Modular Design

Intelligent Battery Management

Distributed Thermal Control

Grid Forming

String PCS

Features

- Three-level topology, up to 99% conversion efficiency, better power quality.
- Supports constant power, constant current, constant voltage control, FCR, VSG, black start and other functions.
- Integrated string-level battery management function, high battery availability.
- Support IEC61850 protocol and have ms-level response capability.
- Intelligent multi-stage fan speed regulation, wide temperature operation capability, no derating at 50°C.
- IP66 protection level, can be installed vertically or horizontally outdoors.



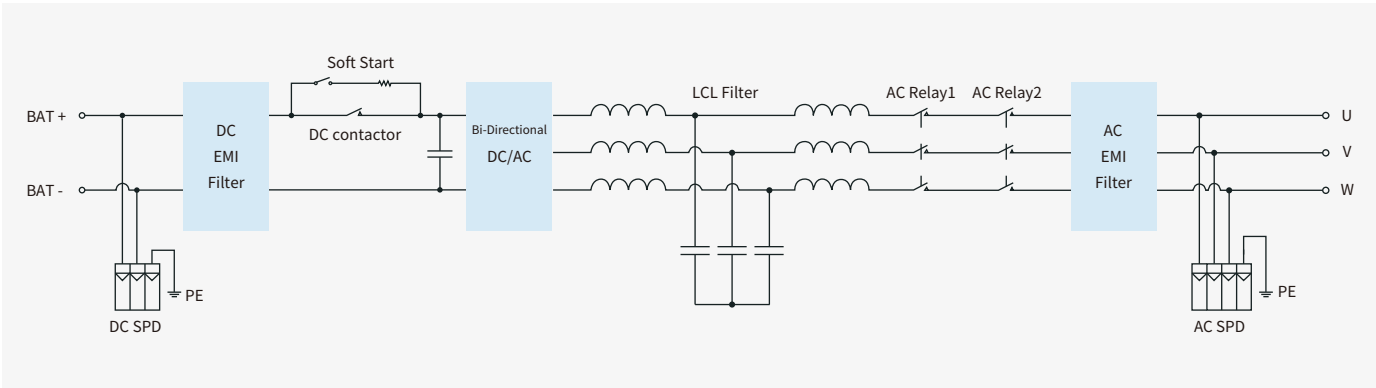
System Technical Specifications

Item	CBAC-125-HA-M	CBAC-140-HA-M-US	EH-0200-HA-M
DC Data			
Maximum DC voltage	1,500V	1,500V	1,500V
Maximum DC power	140kW	157kW	224kW
Maximum DC current	234A	224A	224A
DC voltage range	600~1,100V	700~1,200V	1,000~1,400V
AC Data			
Rated AC power	125kW	140kW	200kW
Maximum AC power	137kW	154kW	220kW
Maximum AC current	198.4A	185.2A	184.1A
Rated grid voltage	400V	480V	690A
Grid voltage range	3W+PE, -15%~+10% (Configurable)		
Rated grid frequency	50/60Hz		
PF adjustable range	-1~+1		
THD (rated power)	<3%		

- Single-stage three-level modularization;
- Multi-branch input to reduce battery series and parallels connection;
- One-to-one Management.

System Features	
Dimensions (W*H*D)	810×845×275mm
weight	98kg
Isolation method	Non-isolated type (external transformer)
Maximum efficiency	99%
Protection level	IP66
Operating temperature range	-40~+60°C
Allowable humidity range	0~100%(No condensation)
Maximum operating altitude	4000m(>3000m Derating)
Cooling method	Temperature controlled forced air cooling
Communication interface	RS485/CAN/Ethernet

Circuit Diagram



SPAC BMS

SPAC BMS technology is developed by UNICOM has been through five times of iteration since 2016, provides strong battery management ability to actively eliminate the imbalanced between battery cells in operation, further guaranteeing the overall availability of BESS in multiple circles while ensuring high safety.

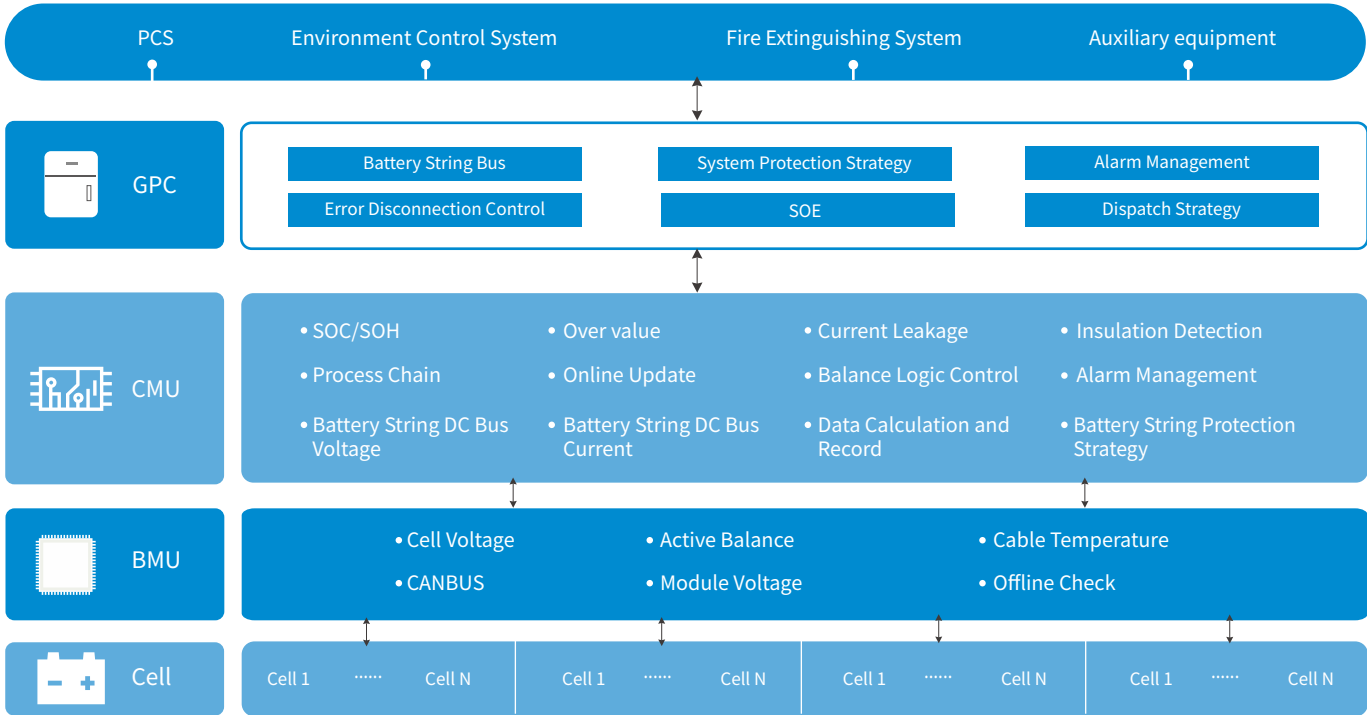
Functions

1. Monitor and protect safety of battery cell

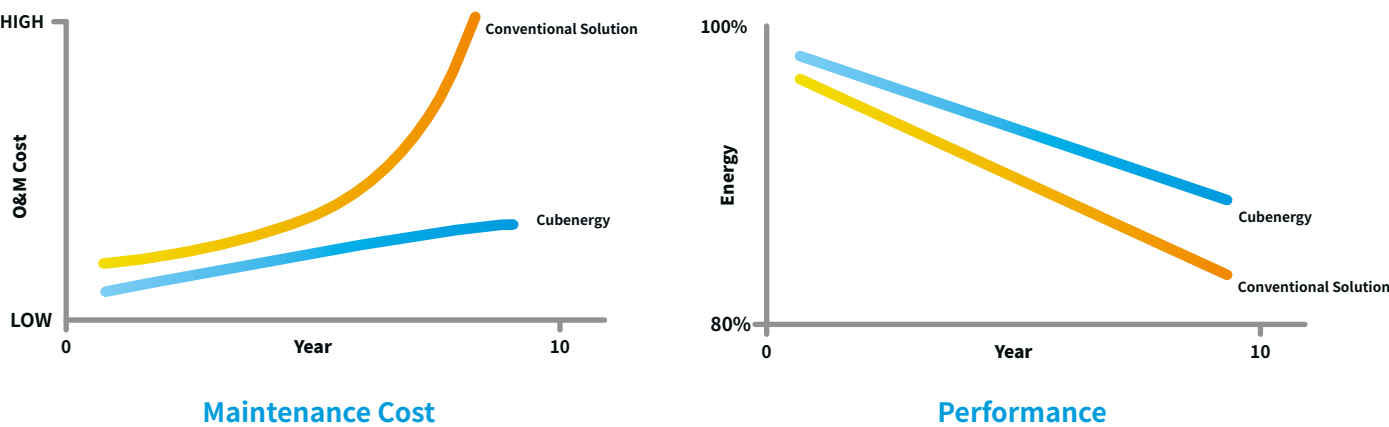
2. Monitor and protect safety of electric system

3. Calculate and manage charging and discharging status
4. Calibrate and manage available energy

5. Optimize balance and manage cell consistency



Advantages



GridPoint Controller



GridPoint Controller (GPC)

an advanced one-stop BESS control and status information and management center.

Power interface
AC400V/DC24V

Communication
Modbus TCP



Super Easy for BESS Power Plant Controller Integrating

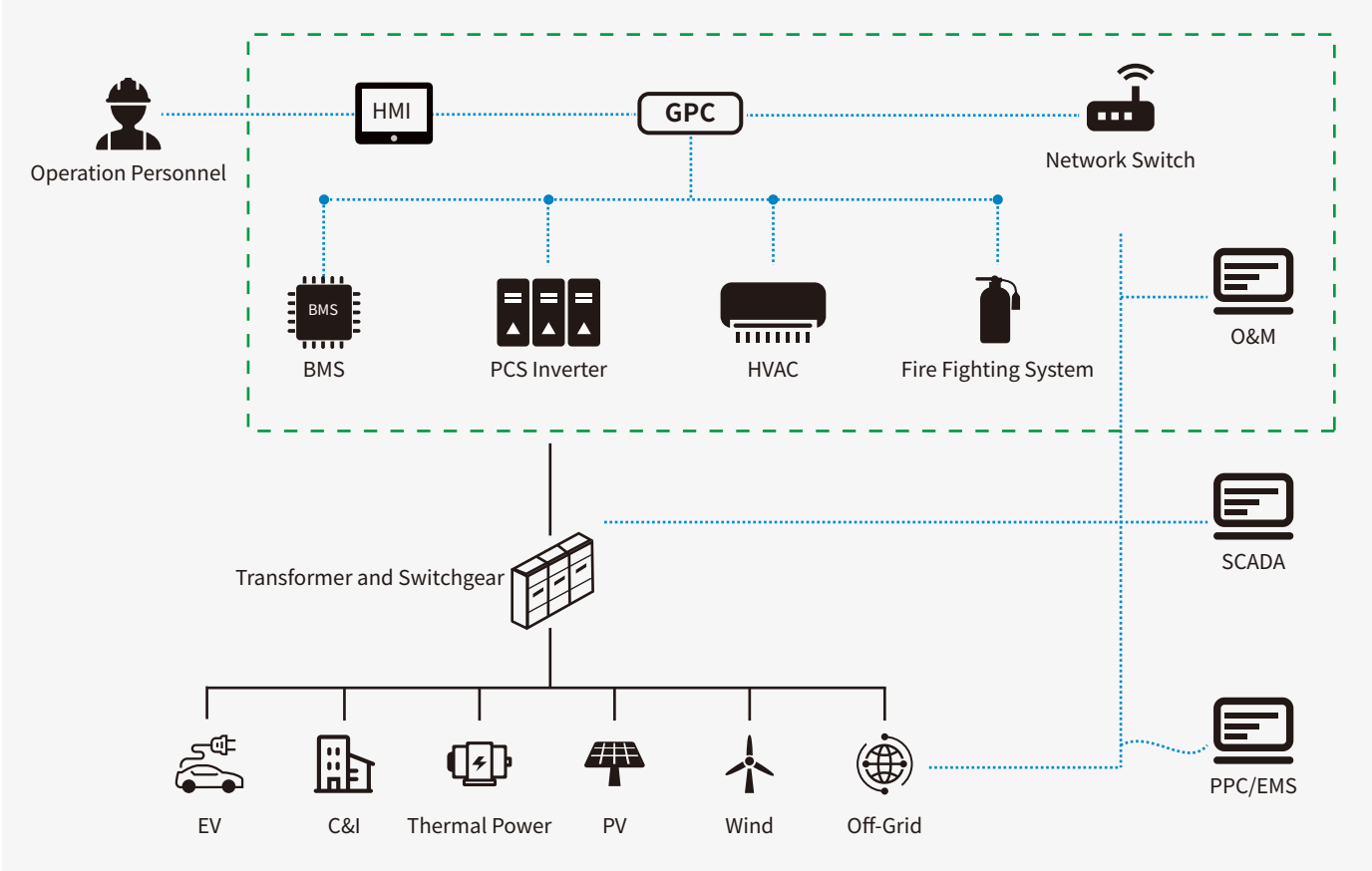


Multi-function to shorten the FAT and SAT with any EMS and SCADA



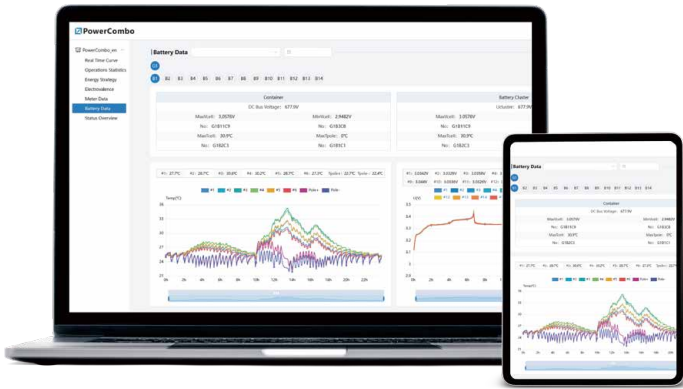
Best tools for on-site operation and maintenance

System Diagram



Operation Platform

Operation Platform is a real-time data monitoring and status alarming platform developed by UNICOM. WeWatt is designed to provide life circle management services to BESS, enable clients to experience easy and intelligent O&M experience. Meanwhile, with 24*7 monitoring and controlling, WeWatt can enhance system safety and improve efficiency in long-term operation.



Functions

- Real-time & Historical Data Monitoring
- Visualized Operation Status Analyzing
- Intelligent Management
- Troubleshooting & Fault Alarm

Features

Data monitoring and analyzing

WeWatt Operation Platform collects real-time status data from batteries to the whole system and analyzes Synchronously, which clients can easily and quickly know about the status of each component from visualized diagrams.

Easy interchange with standard protocol

WeWatt Operation Platform communication is based on standard communication protocol of IEC102, IEC104, and MQTT-The Standard for IoT Messaging, which can easily achieve data interchange.

Service crosswise infinite expansion

WeWatt Operation Platform adopts distributed microservice structure, proceeding crosswise multi-node expansion, ensuring high availability and stability.

Intelligent alarm

WeWatt Operation Platform adopts an intelligent alarm strategy which allows unattended operation. When abnormal data is detected, WeWatt can respond in real-time and send alarm to users by email or text instantly to remind users. Don't need to stick to computer in daily operation basis.

Data safety

Using EU/North America local Amazon server, complying with data security management requirements. API can be created to connect third-party EMS/SCADA databases, extracting the permitted data to analyze and manage.

“ We are committed to letting the data serve device management, providing professional and reliable O&M tools for BESS. Reducing average system downtime and increasing system overall availability through data visualization and daily maintenance. ”



Customer Services

With a decade of hands-on experience in system operation combined with our BMS & power electronics technologies, and know-how in BESS system level, UNICOM takes pride in delivering comprehensive services and greatest effort to support our valued partners.

UNICOM BSS Service

Regular Training

UNICOM provides regular consulting and training services for customers to gain professional guidance and technical knowledge.

Quick Response

With services and inventory network in Europe, and 24-hour hotline, we are committed to a quick and trackable technical response to support your demand and issue.

Preventive Maintenance

Cubenergy is committed to eliminating potential hazards before they arise. Through regular and routine inspection, our engineers can identify potential risks and greatly reduce the chance of system failure or shutdown. By this preventive maintenance and analytical reporting, issues as cell temperature change, cabin temperature drifting, voltage fault, etc, can be prevented in advance to reduce unexpected breakdowns.

Analytical Reporting

The WeWatt cloud-based platform with local Amazon sever in Germany, can monitor real-time operational data from cell to system, providing visualized analysis reports for system SOH assessment, alarm and performance trends, etc.

Grid Code Fine-tune

Under the requirements of different Grid Codes of TSO/DSO in regions, UNICOM's service will dive into interacting with different types of Aggregators. In the most demanding FFR/FCR scene, UNICOM provides fine-tuned service, which enables our customers to maximize their returns while ensuring system availability.

Fast Calibration

Through our innovative SPAC BMS technology and service management, UNICOM provides option of remote auto-fast calibration. Compared to traditional SOC calibration time of 24 hours(Normally calibrated every 1-2 months), UNICOM can shorten the time to within 4 hours and enhance the service frequency, thereby greatly reducing the shutdown time and increasing operational cash flow.

Battery Balance

The fast balancing service based on fast calibration technology can greatly reduce the imbalance of battery capacity, improving the balancing efficiency by more than 90%, compared to traditional balance technology. It also can prevent cell damage caused by overcharge and over-discharge and improve system lifespan.

Europe

Dusseldorf, Germany
Uppsala, Sweden
Lille, France

America

Houston, USA

